NSF Biographical Sketch — SciENcv Entry Worksheet

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Revision 2, 27 Oct 2026: Brian's decisions on the first draft's open items are applied in sections 1–4 and recorded in section 6.

Purpose. NSF requires the Biographical Sketch to be generated in SciENcv (PAPPG NSF 24-1, Chapter II.D.2.h(i); NSPM-33 common form, OMB 3145-0279). This worksheet lists every value to enter, in the order SciENcv asks for it, with the source document for each value. It is a proposal for Brian to review: nothing has been entered into SciENcv, and no existing document was modified.

Sources. BMAnderson CV.docx (saved 30 Sep 2026); Biosketch_BMA.docx (NIH format, 12 Aug 2026); Anderson_CommonFormBiosketch.pdf (NIH common form generated and certified in SciENcv on 26 Feb 2026; the only document that records start months); Crossref, PubMed Central, PyPI and GitHub records checked on 27 Oct 2026.

Legend. **[CONFIRM]** = the documents disagree or are silent; decide before entering. **[DOI NEEDED]** = no persistent identifier could be verified; never guess a DOI. Unflagged values trace directly to a source document.

Form shape. The NSF common form has Identifying Information, Organization and Location, Professional Preparation, Appointments and Positions, Products, and Certification. There is no Synergistic Activities section (that is a separate one-page upload) and no page limit, but content beyond what the form asks for can cause the proposal to be returned without review. Do not enter personal information (home address, phone, etc.).

1. Identifying Information, Organization and Location

Field	Value to enter	Source / notes
Name (Last, First Middle)	Anderson, Brian Mark	CV header; SciENcv biosketch certified 26 Feb 2026.
Persistent identifier (ORCID iD)	https://orcid.org/0000-0002-2748-1444	CLAUDE.md; SciENcv PDF 02/2026. SciENcv fills this from the ORCID account linked to the NCBI login.
Position title	Assistant Professor of Medical Physics	Brian's choice (27 Oct 2026). The CV header says "Clinical Assistant Professor of Medical Physics"; the SciENcv biosketch certified 02/2026 said "Assistant Professor, Medical Physics". Use this wording in all four NSF documents (biosketch, Current & Pending, COA, Synergistic Activities).
Organization (primary)	University of California, San Diego	CV header.
Location (City, State/Province, Country)	San Diego, California, United States	SciENcv uses drop-downs for state and country; type the city.

2. Professional Preparation (reverse chronological by start date)

SciENcv asks, per entry, for organization, location, degree, start date, month/year of receipt, and field of study. Start months are not in the CV (years only) or the NIH biosketch (completion months only); they come from the SciENcv biosketch certified on 26 Feb 2026 and should already be stored in Brian's SciENcv profile. If the new document is created from that existing document, check that they carry over.

Organization	Location	Degree / entry type	Start (MM/YYYY)	Completion (MM/YYYY)	Field of study	Source / notes
University of California, San Diego	San Diego, California, United States	Residency, Medical Physics (no degree; the 02/2026 SciENcv entry used "Resident"; pick the closest option the SciENcv drop-down offers)	07/2021	06/2023 **[CONFIRM: NIH biosketch table says 7/2023; SciENcv 02/2026 says 06/2023]**	Therapeutic Medical Physics (SciENcv 02/2026 used "Therapy Physics")	CV Education: 2021 – 2023 Residency, UCSD. NSF asks for all postdoctoral/fellowship training; the residency is Brian's training entry.
The University of Texas MD Anderson Cancer Center UTHealth Houston Graduate School of Biomedical Sciences	Houston, Texas, United States	Ph.D.	08/2017	05/2021	Medical Physics	CV Education: 2017 – 2021 Ph.D., Medical Physics, MD Anderson; NIH biosketch table 5/2021. Brian confirmed 05/2021 and plain "Medical Physics" (27 Oct 2026). The SciENcv profile from 02/2026 shows 06/2021: correct it there. Dissertation: Improving Treatment of Local Liver Ablation Therapy with Deep Learning and Biomechanical Modeling.

Organization	Location	Degree / entry type	Start (MM/YYYY)	Completion (MM/YYYY)	Field of study	Source / notes
The University of Texas MD Anderson Cancer Center UTHealth Houston Graduate School of Biomedical Sciences	Houston, Texas, United States	S.M.S. (Master of Science)	08/2015	07/2017	Medical Physics	CV Education: 2015 – 2017 S.M.S, Medical Physics, MD Anderson; NIH biosketch table 7/2017. Brian confirmed plain "Medical Physics" (27 Oct 2026); the NIH table's "Therapeutic Medical Physics" is not used.
Georgia Institute of Technology	Atlanta, Georgia, United States	B.S.	08/2010	05/2015	Nuclear and Radiological Engineering	CV Education: 2010 – 2015 B.S, Nuclear Engineering, Highest Honors. NIH biosketch table: BS, 5/2015, Nuclear and Radiological Engineering.

3. Appointments and Positions (reverse chronological by start date)

NSF rule (common form text). "Appointments and positions include any titled academic, professional, or institutional position whether or not remuneration is received, and whether full-time, part-time, or voluntary (including adjunct, visiting, or honorary)." Domestic and foreign professional appointments outside of the primary organization must be listed for a period up to three years before the proposal submission date. Both UNC appointments (Clinical Assistant Professor 2023–2025 and Adjunct Assistant Professor 2025–present) fall inside that window: **do not omit either one.**

3a. Entries to list

Start	End	Title	Organization	Department	Location	Source / notes

Start	End	Title	Organization	Department	Location	Source / notes
2025	present	Assistant Professor of Medical Physics	University of California, San Diego	Department of Radiation Medicine and Applied Sciences	San Diego, California, United States	CV header; NIH biosketch Positions and Employment. Title wording per Brian (27 Oct 2026), matching the position title in section 1. NIH biosketch names the department "Radiation Oncology"; the CV service section uses "Radiation Medicine & Applied Sciences", which is the UCSD department name.
2025	present	Adjunct Assistant Professor	University of North Carolina at Chapel Hill	Department of Radiation Oncology	Chapel Hill, North Carolina, United States	NIH biosketch Positions and Employment; SciENcv 02/2026. OUTSIDE THE PRIMARY ORGANIZATION: must be listed (see rule above). Do not omit.
2025	present	Chair of Technical Innovation and Implementation	University of California, San Diego	Department of Radiation Medicine and Applied Sciences	San Diego, California, United States	CV Professional Service (UCSD Departmental Leadership); NIH biosketch Positions and Scientific Appointments. Titled departmental position: counts as an appointment.
2025	present	Technical Service Chief, Central Nervous System	University of California, San Diego	Department of Radiation Medicine and Applied Sciences	San Diego, California, United States	CV Professional Service; NIH biosketch. Titled departmental position: counts as an appointment.

Start	End	Title	Organization	Department	Location	Source / notes
2023	present	Associate Editor	Medical Physics (journal of the American Association of Physicists in Medicine)	N/A	Alexandria, Virginia, United States (AAPM headquarters)	CV Professional Service (Medical Physics journal); NIH biosketch "Associate Editor: The International Journal of Medical Physics Research and Practice". Titled editorial appointment, unpaid: NSF counts it.
2023	2025	Clinical Assistant Professor	University of North Carolina at Chapel Hill	Department of Radiation Oncology	Chapel Hill, North Carolina, United States	NIH biosketch Positions and Employment; SciENcv 02/2026 (listed as "Assistant Professor, Medical Physics"). Outside the primary organization and within three years of submission: must be listed.
2021	2023	Clinical Resident, Medical Physics	University of California, San Diego	Department of Radiation Medicine and Applied Sciences	San Diego, California, United States	NIH biosketch Positions and Employment; CV Education. Also appears under Professional Preparation. The 02/2026 SciENcv biosketch listed it only there; listing it in both places is allowed. Kept here as a titled training position.
2015	2021	Graduate Student Researcher	The University of Texas MD Anderson Cancer Center	Department of Imaging Physics	Houston, Texas, United States	NIH biosketch Positions and Employment ("Graduate Student Researcher, MDA Cancer Center, Houston, TX").

Start	End	Title	Organization	Department	Location	Source / notes
2014	2014	AAPM Summer Undergraduate Fellow	St. Joseph Hospital	Radiation Oncology	Eureka, California, United States	NIH biosketch Positions and Employment ("2014 AAPM Fellowship, St. Joseph's Hospital, Eureka, CA"); CV Grants and Fellowships 2014.
2013	2014	Research Assistant	Georgia Institute of Technology	Nuclear and Radiological Engineering	Atlanta, Georgia, United States	NIH biosketch Positions and Employment ("2013-2014 Research Assistant, Georgia Institute of Technology").

3b. Other titled roles and memberships (Brian's call)

NSF counts titled positions. Brian decided on 27 Oct 2026 that his committee memberships go in the Synergistic Activities document, not here; the two remaining optional titled roles are his call.

Start	End	Title / role	Organization	Recommendation
2026	2027	Search Committee Member (medical physicist search)	University of California, San Diego	Omit: internal committee duty, not a titled appointment.
2026	present	Member, QUANTEC 2 Working Group (Q2WG)	American Association of Physicists in Medicine (AAPM)	Omit here; list under Synergistic Activities (Brian, 27 Oct 2026).
2024	present	Member, Machine Intelligence Subcommittee	AAPM	Omit here; list under Synergistic Activities (Brian, 27 Oct 2026).
2023	present	Member, Medical Physics Board of Associate Editors	AAPM	Omit here (redundant with the Associate Editor appointment in 3a); may be mentioned under Synergistic Activities.
2021	present	Member, Task Group 263 (Standardizing Nomenclatures in Radiation Oncology)	AAPM	Omit here; list under Synergistic Activities (Brian, 27 Oct 2026). End date is "Present"; the CV's Major Contributions entry (2021–2025) is being corrected in a separate session.
2021	2023	Chief Resident	University of California, San Diego	Optional titled role (CV Clinical Projects); the 02/2026 SciENcv omitted it. Add to 3a only if Brian wants it listed.
2020	2021	Ad hoc Associate Editor	Medical Physics (AAPM)	Optional titled editorial role (CV Professional Service). Add to 3a only if Brian wants it listed.
2019	present	Manuscript Reviewer	Medical Physics (AAPM)	Omit: reviewing is not a titled appointment.

Start	End	Title / role	Organization	Recommendation
2015 / 2017	present	Member, AAPM; Member, Medical Physicists for World Benefit	AAPM; MPWB	Omit: society memberships are not appointments.

4. Products

Scope. The split below was drawn up for a proposal on AI / deep learning for radiation oncology (auto-segmentation, deformable registration, clinical software and data infrastructure), and Brian confirmed it on 27 Oct 2026. If a later proposal is about something else, swap items between 4a, 4b and the bench list in 4c. NSF allows up to five products in each list; only these ten are used in review.

Entry rules. Products must be citable and accessible: publications, software, datasets, patents and the like; unpublished documents, invited lectures and "additional lists" are not allowed. Each entry needs authors ("et al." allowed after some authors), title, venue with volume/issue/pages where applicable, release date, and a DOI or URL. In SciENcv, add products from My Bibliography / ORCID where they exist; everything else is typed in manually. Citations can be reordered by dragging.

4a. Products most closely related to the proposed project (max 5)

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
A1	Anderson BM, Wahid KA, Brock KK	Simple Python Module for Conversions Between DICOM Images and Radiation Therapy Structures, Masks, and Prediction Arrays	Practical Radiation Oncology, 11(3), 226–229	05/2021	https://doi.org/10.1016/j.prro.2021.02.003 PMCID PMC8102371	In PubMed: should appear in My Bibliography / ORCID. CV lists the submitted title ("Simple Python Module for Dicom and RT …") and 02/2021; use the published title.

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
A2	Anderson BM	DicomRTTool: open-source Python library converting DICOM images, RT structure sets and dose grids to NumPy/SimpleITK arrays, and model predictions back to DICOM-RT structures (software)	GitHub repository brianmanderson/Dicom_RT_and_Images_to_Mask; PyPI package DicomRTTool (first release 09/2020; current v6.2.0, 07/2026)	09/2020 (first release)	https://github.com/brianmanderson/Dicom_RT_and_Images_to_Mask https://pypi.org/project/DicomRTTool/	CV Major Contributions. Software is an allowed NSF product (citable, accessible). Not in ORCID: enter manually in SciENcv using the non-journal citation type. Checked 27 Oct 2026: 108 GitHub stars, 32 forks. Usage figures are optional context, not part of the citation.
A3	Anderson BM, Padilla L, Ryckman JM, Covington E, Hong DS, Woods K, Katz MS, Zuhour R, Estes C, Moore KL, Bojechko C	Open RT Structures: A Solution for TG-263 Accessibility	International Journal of Radiation Oncology, Biology, Physics, 118(3), 859–863	03/2024	https://doi.org/10.1016/j.ijrobp.2023.09.041	Listed in the 02/2026 SciENcv biosketch with this citation. CV gives 06/2024 and omits Zuhour R; Crossref and the 02/2026 SciENcv give the March 2024 issue and 11 authors (CV fix is a separate task).
A4	Anderson BM, Lin EY, Cardenas CE, Gress DA, Erwin WD, Odisio BC, Koay EJ, Brock KK	Automated Contouring of Contrast and Noncontrast Computed Tomography Liver Images With Fully Convolutional Networks	Advances in Radiation Oncology, 6(1), 100464	01/2021 (online 05/2020)	https://doi.org/10.1016/j.adro.2020.04.023 PMCID PMC7807136	In PubMed. CV lists 05/2020 (online date); SciENcv 02/2026 used Jan–Feb 2021.

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
A5	Anderson B, Repka M, Gao Y, Byrd E, McGurk R, Das S, Dooley J, Arcot R, Bhattacharya S, Adapa K, Wright J, Marks LB, Mazur LM	Using a Deep Learning System to Delineate the Elective Pelvic Nodal Volumes in Patients with Prostate Cancer: A Tool Aimed to Facilitate Peer Review	Advances in Radiation Oncology, article 102219; volume/issue **[CONFIRM: not yet assigned in Crossref on 27 Oct 2026]**	09/2026 (Crossref); CV lists 07/2026	https://doi.org/10.1016/j.adro.2026.102219	Absent from ORCID (2022+ gap): enter manually or push to ORCID first. CV author list omits Dooley J (Crossref lists 13 authors).

4b. Other significant products (max 5)

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
B1	Odisio BC, Albuquerque J, Lin YM, Anderson BM, O'Connor CS, Rigaud B, et al. (30 authors)	Software-based versus visual assessment of the minimal ablative margin in patients with liver tumours undergoing percutaneous thermal ablation (COVER-ALL): a randomised phase 2 trial	The Lancet Gastroenterology & Hepatology, 10(5), 442–451	05/2025	https://doi.org/10.1016/S2468-1253(25)00024-X	Phase 2 RCT (NCT04083378) built on Brian's dissertation registration and segmentation tools (CV Major Contributions). Published title reads "randomised phase 2 trial"; the CV adds "controlled". Crossref spells the 16th author Shah KY (CV: Shan).
B2	Anderson BM, Lin YM, Lin EY, Cazoulat G, Gupta S, Jones AK, Odisio BC, Brock KK	A novel use of biomechanical model-based deformable image registration (DIR) for assessing colorectal liver metastases ablation outcomes	Medical Physics, 48(10), 6226–6236	10/2021 (online 08/2021)	https://doi.org/10.1002/mp.15147 PMCID PMC9380122	In PubMed; listed in the 02/2026 SciENcv biosketch. CV lists 07/2021.

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
B3	Anderson B, Moore L, Bojechko C	Rapid in vivo EPID image prediction using a combination of analytically calculated attenuation and AI predicted scatter	Medical Physics, 52(2), 1058–1069	02/2025 (online 11/2024)	https://doi.org/10.1002/mp.17549	CV Papers 02/2025. Likely absent from ORCID: enter manually.
B4	Anderson BM, Bojechko C	DICOM attribute manipulation tool: Easily change frame of reference, series instance, and SOP instance UID	Journal of Applied Clinical Medical Physics, 26(6), e70104	06/2025 (online 04/2025)	https://doi.org/10.1002/acm2.70104	Listed in the 02/2026 SciENcv biosketch. CV title says “Study Instance UID”; the published title says “SOP instance UID”.

#	Authors (plain text)	Title	Venue, volume(issue), pages	Date	DOI / URL	Notes
B5	Anderson BM	keras.ops.image.reconstruct_patches for Keras 3: the inverse of extract_patches, reassembling non-overlapping 2D or 3D patches of model predictions into full-resolution images and volumes (software contribution)	keras-team/keras (GitHub), merged to master via pull request #22984 on 16 July 2026; follow-ups #23267 (channels_first support, 20 July 2026) and #23319 (output_size auto-inference, 4 September 2026)	07/2026 (merged); in no tagged release as of 27 Oct 2026	https://github.com/keras-team/keras/pull/22984	CV Major Contributions. Enter manually (non-journal type). Cite the pull request, not a version: a separate session confirmed on 27 Oct 2026 that no tagged Keras release contains it yet (latest v3.15.1, 29 July 2026; master carries version 3.16.0) and no keras.io page exists. Re-check for a v3.16 tag before certifying. The ReconstructPatches2D/3D layers named in the CV and in the pull-request title were removed before merge and are not in Keras; the CV entry needs correcting (separate task). PR #23930 (overlapping patches) is still open and must not be described as merged.

4c. Bench: verified alternates to swap in

#	Citation (plain text)	DOI / URL	Notes
C1	Anderson BM, Rigaud B, Lin YM, Jones AK, Kang HC, Odisio BC, Brock KK. Automated segmentation of colorectal liver metastasis and liver ablation on contrast-enhanced CT images. Frontiers in Oncology, 12, 886517, 08/2022.	https://doi.org/10.3389/fonc.2022.886517 PMCID PMC9403767	Was related product #2 in the 02/2026 SciENcv biosketch. Journal is Frontiers in Oncology (Radiation Oncology section); the CV says "Frontiers in Radiation Oncology".
C2	Rigaud B, Anderson BM, Yu ZH, Gobeli M, Cazoulat G, Söderberg J, et al. Automatic Segmentation Using Deep Learning to Enable Online Dose Optimization During Adaptive Radiation Therapy of Cervical Cancer. International Journal of Radiation Oncology, Biology, Physics, 109(4), 1096–1110, 03/2021.	https://doi.org/10.1016/j.ijrobp.2020.10.038	CV lists 10/2020 with the abstract's title. Do not cite 10.1016/j.ijrobp.2020.07.2573 : that is the ASTRO 2020 abstract, not the paper.
C3	Cardenas CE, Yang J, Anderson BM, Court LE, Brock KK. Advances in Auto-Segmentation. Seminars in Radiation Oncology, 29(3), 185–197, 07/2019.	https://doi.org/10.1016/j.semradonc.2019.02.001	Review article; good "other significant" candidate if a segmentation review is wanted.
C4	Lin YM, Paolucci I, O'Connor CS, Anderson BM, Rigaud B, Fellman BM, Jones KA, Brock KK, Odisio BC. Ablative Margins of Colorectal Liver Metastases Using Deformable CT Image Registration and Autosegmentation. Radiology, 307(2), e221373, 04/2023.	https://doi.org/10.1148/radiol.221373	Clinical validation of the dissertation tools.
C5	Covington EL, Suresh K, Anderson BM, Barker M, Dess K, Price JG, et al. Perceptions on and roadblocks to implementation of standardized nomenclature in radiation oncology: A survey from TG-263U1. Journal of Applied Clinical Medical Physics, 25(6), e14359, 06/2024.	https://doi.org/10.1002/acm2.14359	Pairs with Open RT Structures for a standardization-themed proposal.
C6	Floyd W, Wahid KA, Anderson B, Knäusl B, Moreno AC. Large language models for scalable standardization in radiation oncology. Physics and Imaging in Radiation Oncology, 38, 100960, 03/2026.	https://doi.org/10.1016/j.phro.2026.100960	Absent from ORCID; enter manually if used.
C7	Gao Y, Wells D, Bhattacharya S, Repka M, Lavrova L, Anderson B, et al. Deep Learning-Based Pelvic Vessel Auto-Segmentation for Standardized Lymph Node Delineation in Prostate Cancer Radiotherapy. Practical Radiation Oncology, 09/2026; volume, issue and pages **[CONFIRM: not yet assigned in Crossref]**.	https://doi.org/10.1016/j.prro.2026.09.009	Companion to A5. Absent from ORCID.

#	Citation (plain text)	DOI / URL	Notes
C8	Anderson BM, Brock KK. Artificial intelligence-based image registration and segmentation. In: Wang Y, Qi XS (eds). Artificial Intelligence in Adaptive Radiation Therapy. IOP Publishing, 11/2025, chapter 9, pp. 9-1 to 9-31. ISBN 9780750361194.	https://doi.org/10.1088/978-0-7503-6119-4ch9	Book chapter, published (Crossref record checked 27 Oct 2026). The CV lists it as "Chapter 9: Imaging Registration and Auto-Segmentation" with no publisher or date; the published chapter title is shown here.
—	Not eligible: the medRxiv preprint (Castelo et al., 02/2026) until accepted; conference abstracts and presentations; invited talks.	—	NSF: "unpublished documents not yet submitted for publication, invited lectures, and additional lists of products" are unacceptable.

5. Certification

The certification text lives inside SciENcv and is not typed in. After the sections above are complete, click Download PDF; SciENcv shows the certification ("I certify that the information provided is current, accurate, and complete … I am not a party to a malign foreign talent recruitment program …") and Brian must click CERTIFY from his own NCBI account. A delegate can draft the document but cannot certify it. The certification date must be within 12 months of the proposal submission date; the PDF is then uploaded to the proposal in Research.gov.

6. Decisions and open items

6a. Decided by Brian (27 Oct 2026)

- Position title: "Assistant Professor of Medical Physics", used in all four NSF documents.
- Ph.D. completion 05/2021; field of study "Medical Physics" for both the S.M.S. and the Ph.D. The SciENcv profile from 02/2026 shows 06/2021 for the Ph.D. and must be corrected when the new document is built.
- Department names in section 3a and the AAPM location for the Associate Editor entry are correct as written.
- AAPM committee memberships (QUANTEC 2 Working Group, Machine Intelligence Subcommittee, TG-263, Board of Associate Editors) go in Synergistic Activities, not in Appointments.
- TG-263 membership end date is "Present"; the CV's Major Contributions entry (2021–2025) is being corrected in a separate session.
- Product split in 4a/4b confirmed. The Keras release question was investigated separately: B5 is in no tagged release as of 27 Oct 2026, so the citation points at the merged pull request.
- ORCID gap acknowledged: A2, A5, B3, B5 (and C6, C7, C8 if used) will not import; they are entered manually in SciENcv or pushed to ORCID first.

6b. Still open

- Residency completion month: 06/2023 (SciENcv 02/2026) or 07/2023 (NIH biosketch table). **[CONFIRM]**
- Whether to add the two optional titled roles in 3b (Chief Resident 2021–2023; Ad hoc Associate Editor 2020–2021) to the Appointments list.
- Volume/issue/pages for A5 (Advances in Radiation Oncology 2026) and C7 (Practical Radiation Oncology 2026): not yet assigned by the journals; check again before certifying.
- Before certifying, re-check whether a Keras v3.16 tag exists and contains merge commit bbd36de (PR #22984); if so, add the version and date to B5.
- The CV's Major Contributions entry credits ReconstructPatches2D/3D layers that were dropped from PR #22984 before merge; only the reconstruct_patches op is in Keras. Correct the CV entry (separate /update-cv task, Brian's approval).
- Certify in SciENcv from Brian's own account; the certification date must fall within 12 months of proposal submission. The 02/2026 certified PDF in this repo is the NIH common form, not the NSF document, and cannot be submitted to NSF.
- CV citation discrepancies found while verifying DOIs are being corrected in a separate session; they do not affect the citations in section 4, which already use the published records.